

Task : Basic Linux Commands

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1. Objective -

The objective of this task was to establish a remote connection to a target machine using (Secure Shell) SSH and upon successful login, to identify the currently logged-in user and the current working directory on that machine.

2. Prerequisites – What is SSH?

SSH (Secure Shell) is a network protocol used to securely access and manage a remote computer over an unsecured network. Unlike older protocols such as Telnet or FTP, which send data(including usernames and passwords) in plain text, SSH encrypts the entire session. This means that even if someone intercepts the traffic between the client and the server, they cannot read the credentials or commands being exchanged.

SSH is the standard method that system administrators, developers, and security professionals use to manage remote Linux servers, which is why this exercise forms a foundational part of cybersecurity training.

3. Step-by-step process

Step 1: Initiating the SSH Connection

I opened a terminal in my kali linux machine and connected to the target machine using the credentials provided in the tryhackme room.

Now Use the Below Creds to Get Started

Credentials			
Only needed if you are using your own machine.			
Username	Password	IP address	Connection via
 student	 student123	 MACHINE_IP	 SSH <code>ssh student@MACHINE_IP</code>

The command I used to connect to the target machine was :

```
ssh student@192.168.29.188
```

- `ssh` - invokes the SSH client program.
- `student` (username) - the account I was authenticating as on the remote machine.
- `192.168.29.188` (target-ip) - the address of the machine I was connecting to.

Upon running this, I was prompted to enter the password for the account

Step 2: Confirming Successful Login

Once authenticated, the terminal prompt changes from `(kali@kali) - [~]` (local Kali machine) to `student@hacking:~$` (the remote target) after the SSH connection succeeds.

The terminal showing “Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.8.0-101-generic x86_64)” and the change of terminal prompt were the visual indicators that the connection was successful and that I was now operating inside a shell on the target machine rather than my local machine.

Step 3: Verifying the Logged-in User

To confirm which username I was logged in as, I ran the command :

```
whoami
```

What it does: `whoami` prints the username of the account currently running the shell session.

Why it matters: After connecting via SSH, it's important to verify identity before running any further commands — especially on shared or multi-user systems, where it's easy to assume you're logged in as one user when you're actually operating as another.

Output observed: `student`

Interpretation: This confirmed that the SSH session had authenticated me as the intended user account `student`.

Step 4: Verifying the Current Directory

To confirm the current working directory , I ran the command :

```
pwd
```

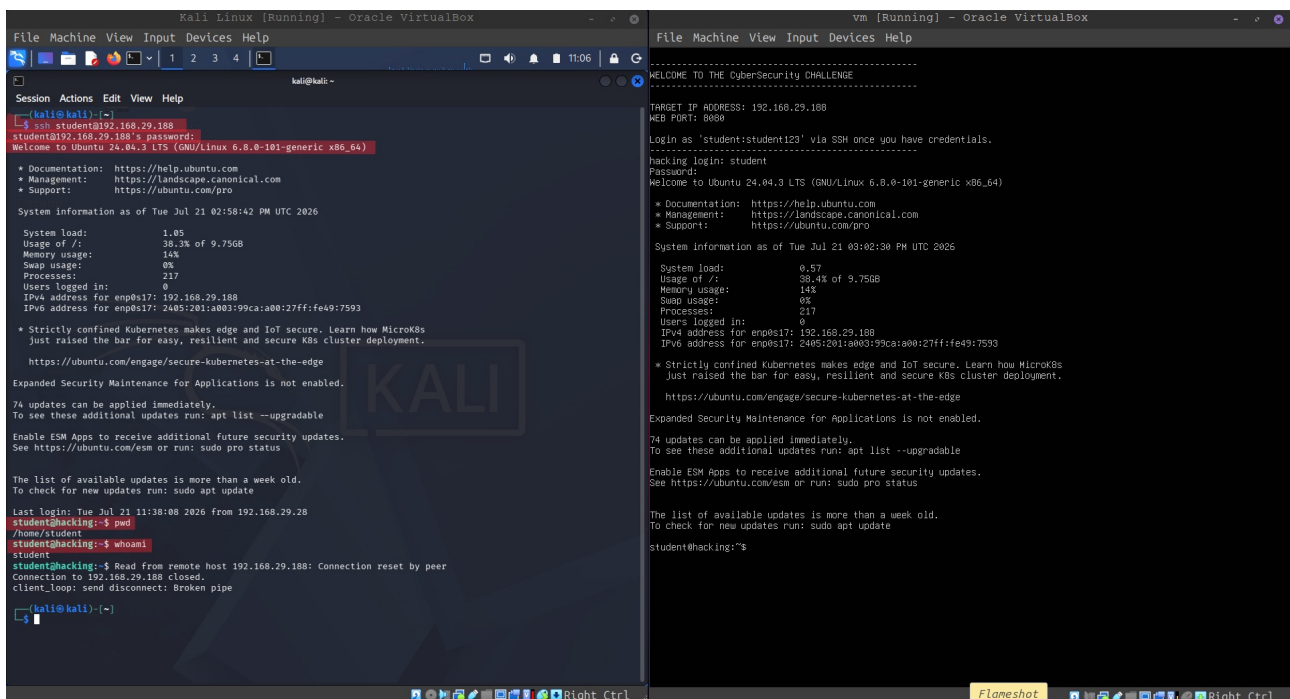
What it does: `pwd` (print working directory) outputs the absolute path of the directory the shell is currently present in.

Why it matters: When you log in via SSH, you're placed by default into the account's home directory. Confirming this path verifies that the session started in the expected location, which matters before running any file operations, scripts, or navigating the filesystem, running a command in the wrong directory can have unintended consequences.

Output observed: /home/student

Interpretation: The output confirmed that I landed in the home directory of the account immediately after login.

Step 5: Screenshot Evidence



4. Conclusion

This task helped me understand the basic but essential workflow of connecting to a remote Linux machine via SSH and verifying one's identity and location on that system using `whoami` and `pwd` which are foundational to safe system administration and cybersecurity practice.